

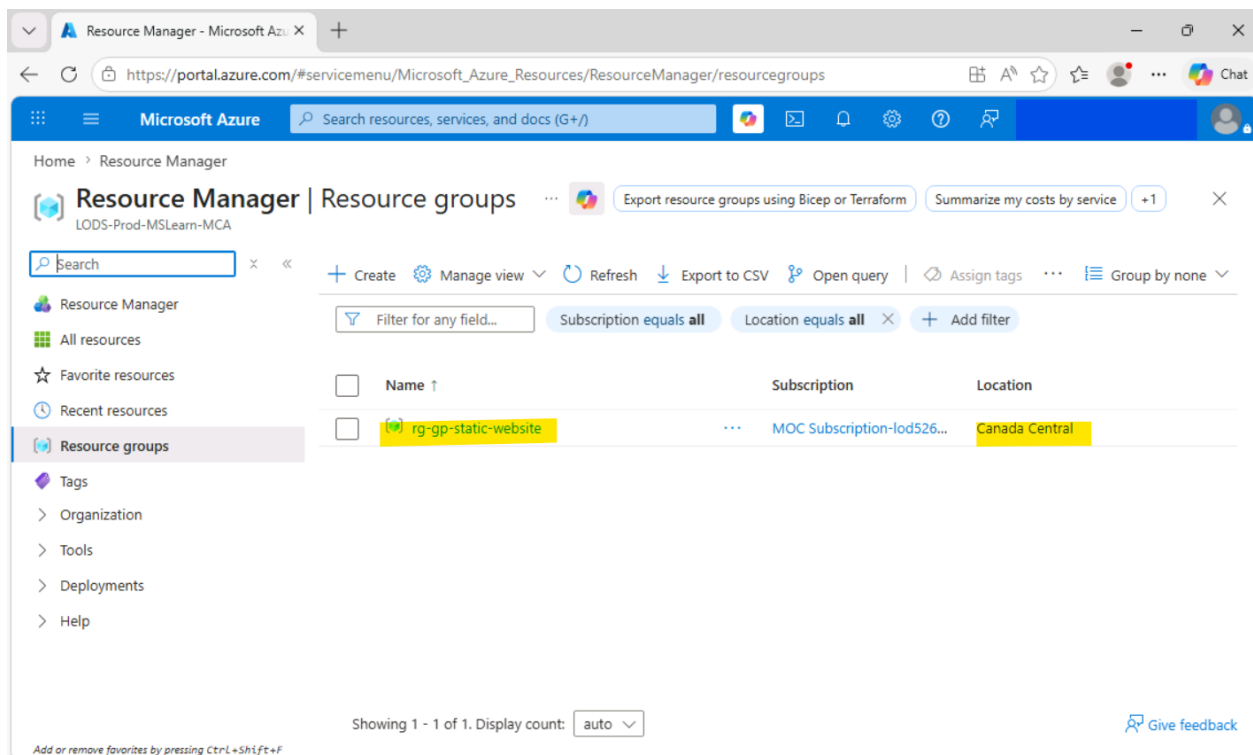
Deploy a static website with Azure Blob-Storage

Project Overview

This lab demonstrates how to deploy a static website using Azure Blob Storage. Azure Storage Accounts can host static web content such as HTML, CSS, JavaScript, and image files without requiring a dedicated web server. By enabling the Static Website feature, Azure automatically creates a **\$web** container and provides a public endpoint URL for accessing the website. In this exercise, a simple website and a custom 404 error page are uploaded to Azure Blob Storage and tested through the generated endpoint.

Prepare the environment

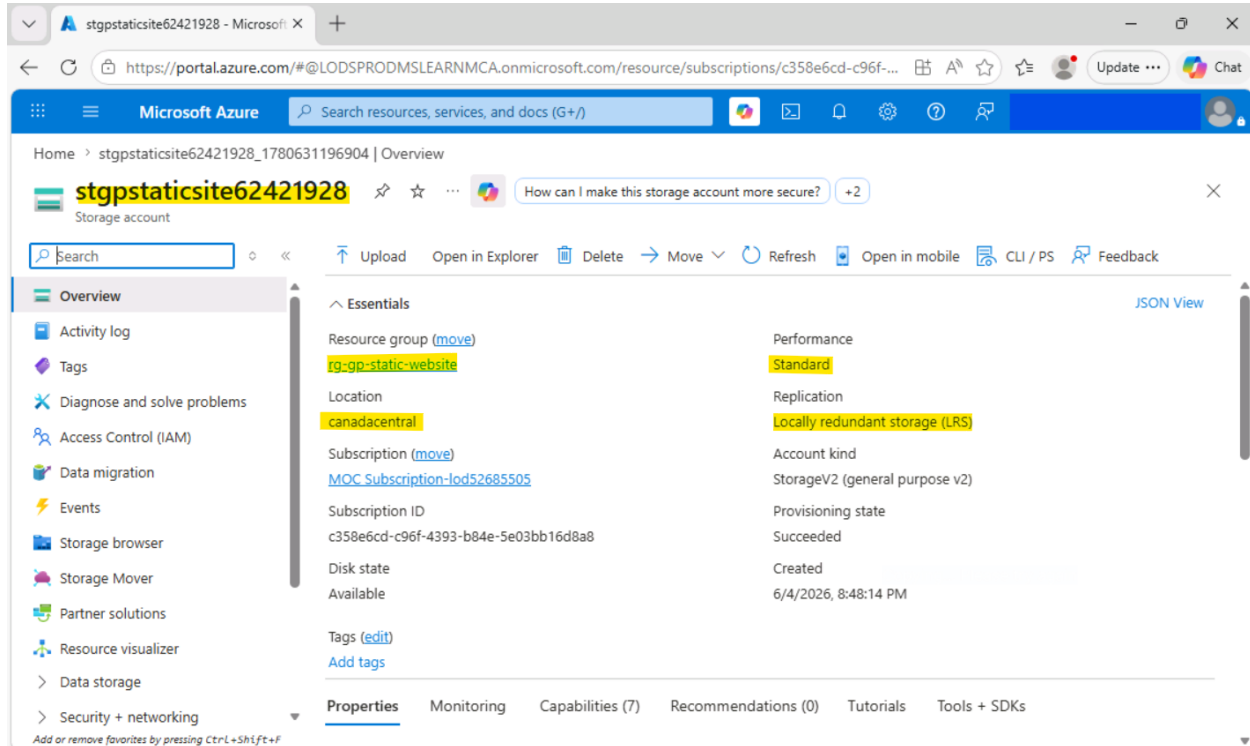
Create a resource group to organize all the resources for this project together, which makes cleanup easier when you finish.



Create a Resource Group named 'rg-gp-static-website' in Canada Central Region.

Create the Storage account

Create a general-purpose storage account that will host your website files. Azure Blob Storage can serve static HTML, CSS, and image files directly to browsers without requiring a web server.

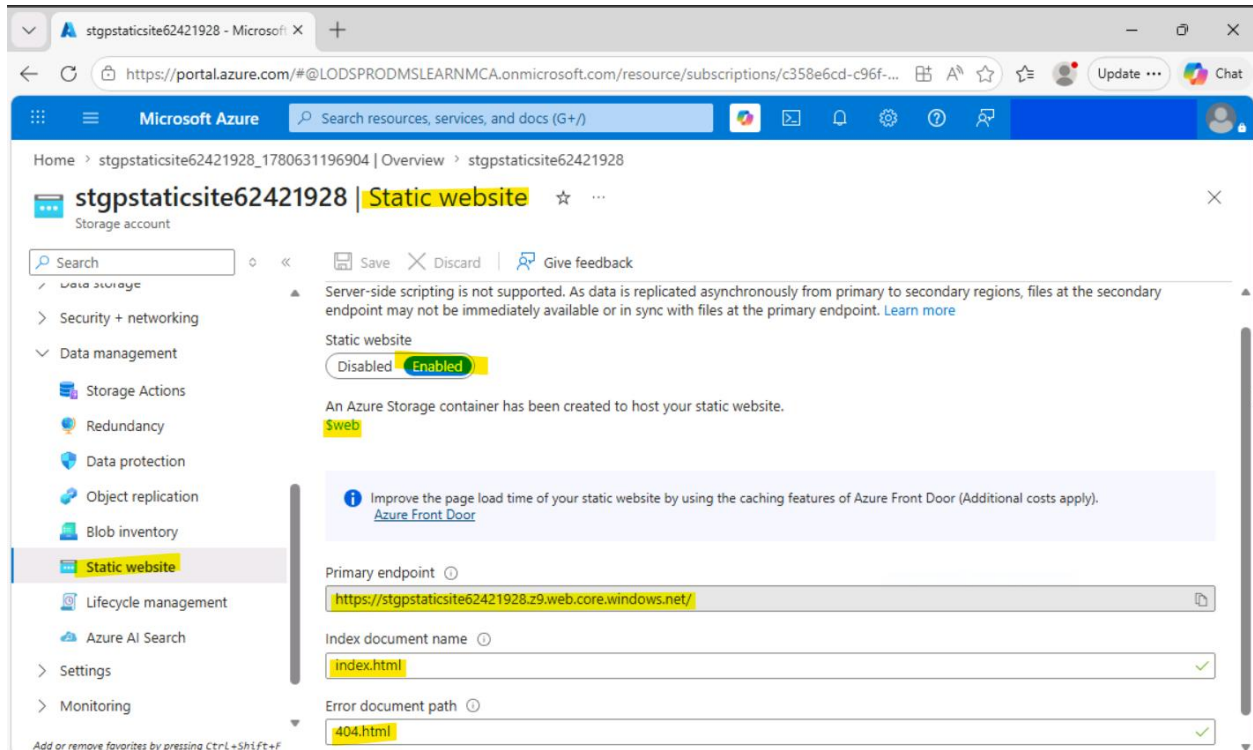


Create a Blob storage named 'stgpstaticsite62421928' in the Resource Group.

- For Resource Group, select 'rg-gp-static-website'.
- For Region, select same as the Resource Group.
- For Primary Service, select **Azure Blob Storage or Azure Data Lake Storage**.
- For Performance, select **Standard**.
- For **Redundancy**, select **Locally-redundant storage (LRS)**.

Enable static website hosting

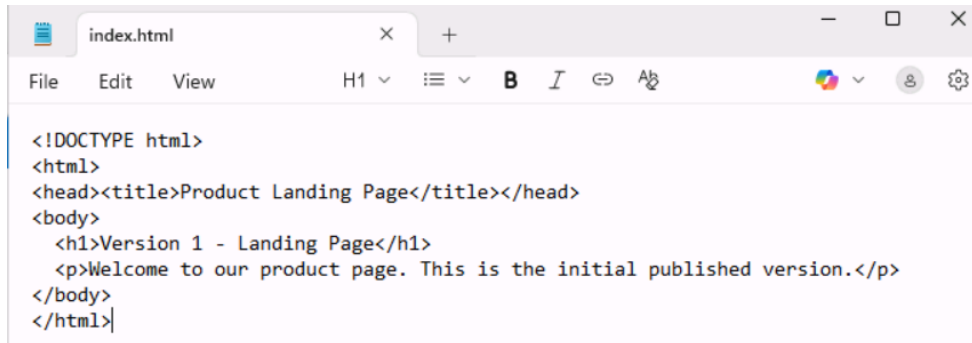
Turn on the built-in static website feature. This creates a special container named \$web where you upload your site files and provides a public URL that serves those files to visitors.



Enable 'Static Website' in the Blob Storage. Set 'index.html' as index document name and '404.html' as Error document path.

Create the HTML file locally

Create a simple HTML file on your computer that will serve as the initial version of your website. This file represents the content your team wants to publish.



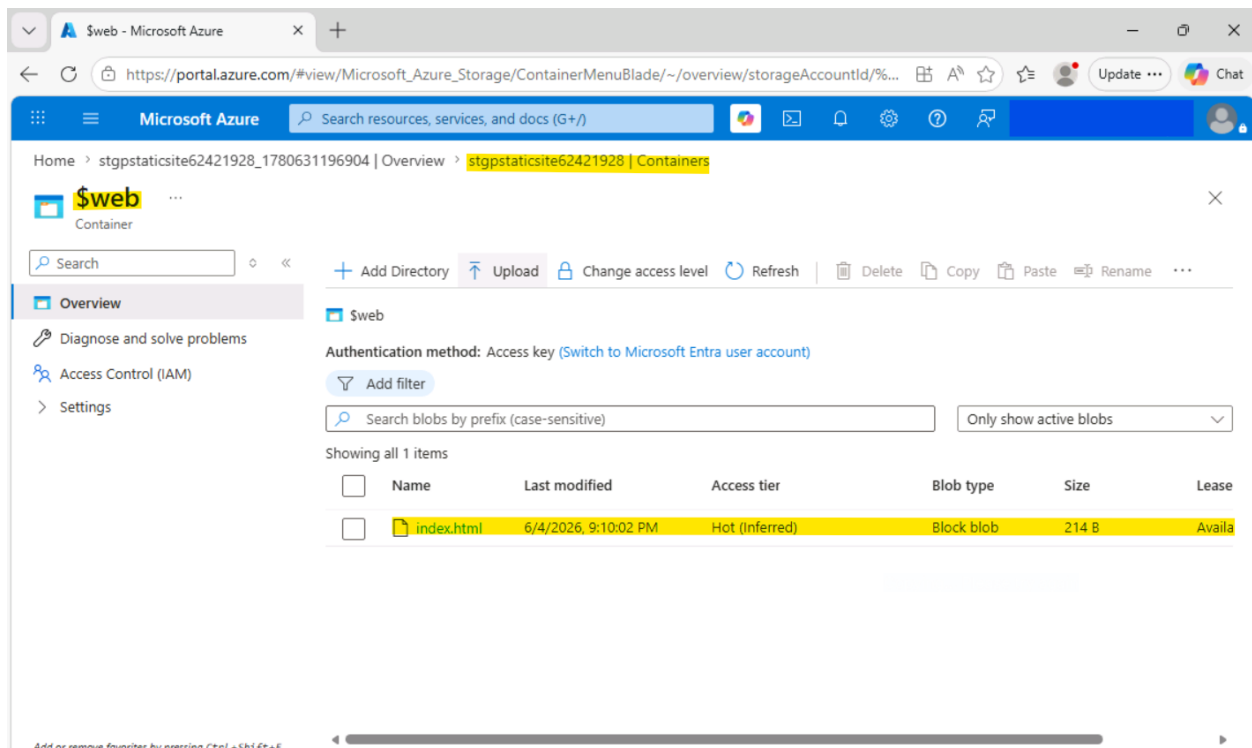
```
<!DOCTYPE html>
<html>
<head><title>Product Landing Page</title></head>
<body>
  <h1>Version 1 - Landing Page</h1>
  <p>Welcome to our product page. This is the initial published version.</p>
</body>
</html>
```

Create a simple index.html file locally using notepad.

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Upload the file to the \$web container

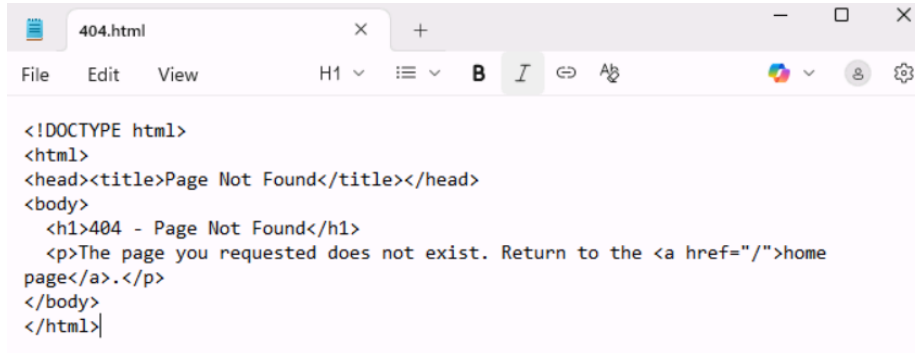
Upload your HTML file to the special \$web container that Azure created when you enabled static website hosting. Files in this container are automatically served to visitors who browse to your site endpoint.



Upload the HTML file to the \$web container.

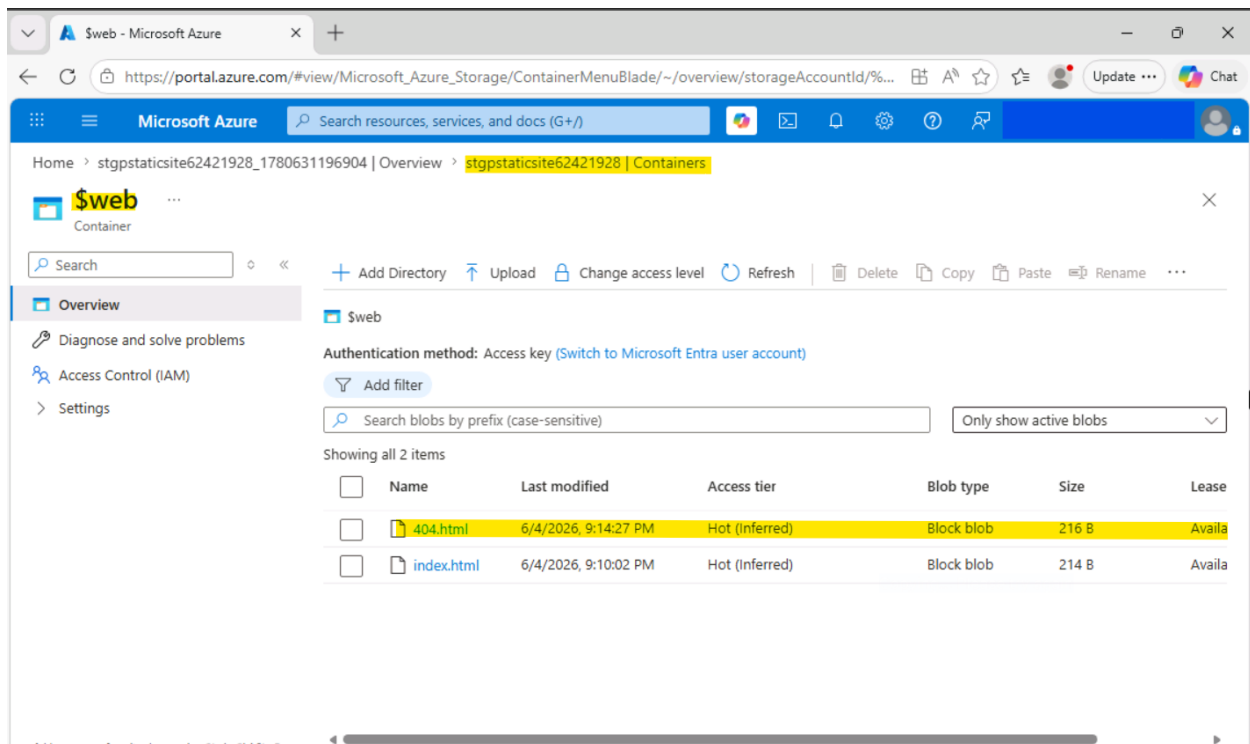
Create and upload a custom error page

Create a custom 404 error page that visitors see when they browse to a page that doesn't exist. Without this file, visitors get a generic XML error, which looks unprofessional.



```
<!DOCTYPE html>
<html>
<head><title>Page Not Found</title></head>
<body>
  <h1>404 - Page Not Found</h1>
  <p>The page you requested does not exist. Return to the <a href="/">home
page</a>.</p>
</body>
</html>
```

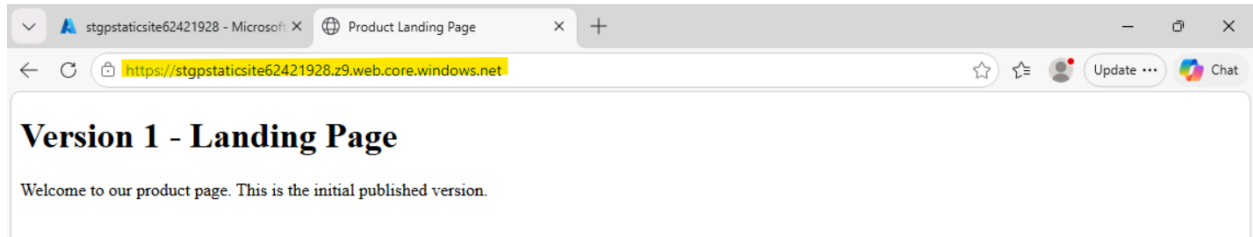
1. Create a 404error.html page.



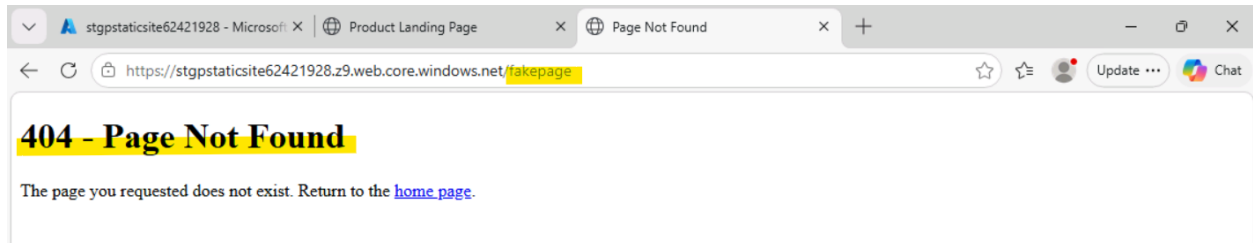
2. Upload the 404 file to the \$web container.

Verify the live site and error page

Open the public endpoint in a browser to confirm your website is live, and then test the custom error page. This validates that static website hosting is correctly configured for both normal pages and missing pages.



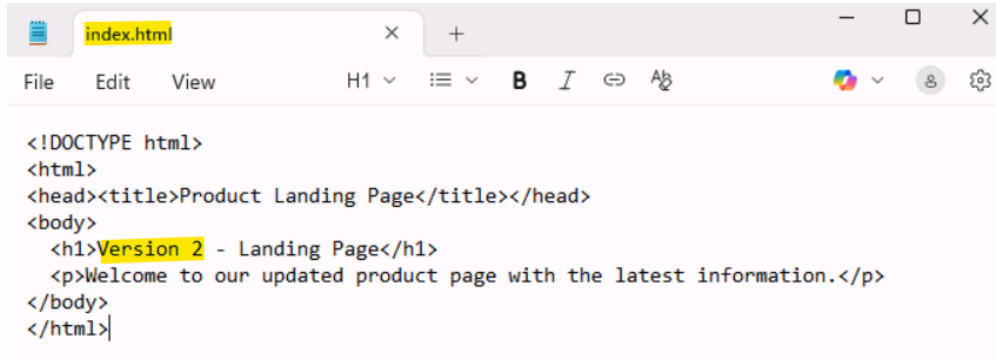
1. Go to the storage account and select Static Website under Data management. Copy the endpoint URL and paste it to the new browser.



2. Add '/fakepage' at the end of the endpoint URL to see if the 404 page works.

Create the updated HTML file

Prepare a new version of the page content. In a real scenario, this represents a content update from the marketing team that needs to replace the current live page.



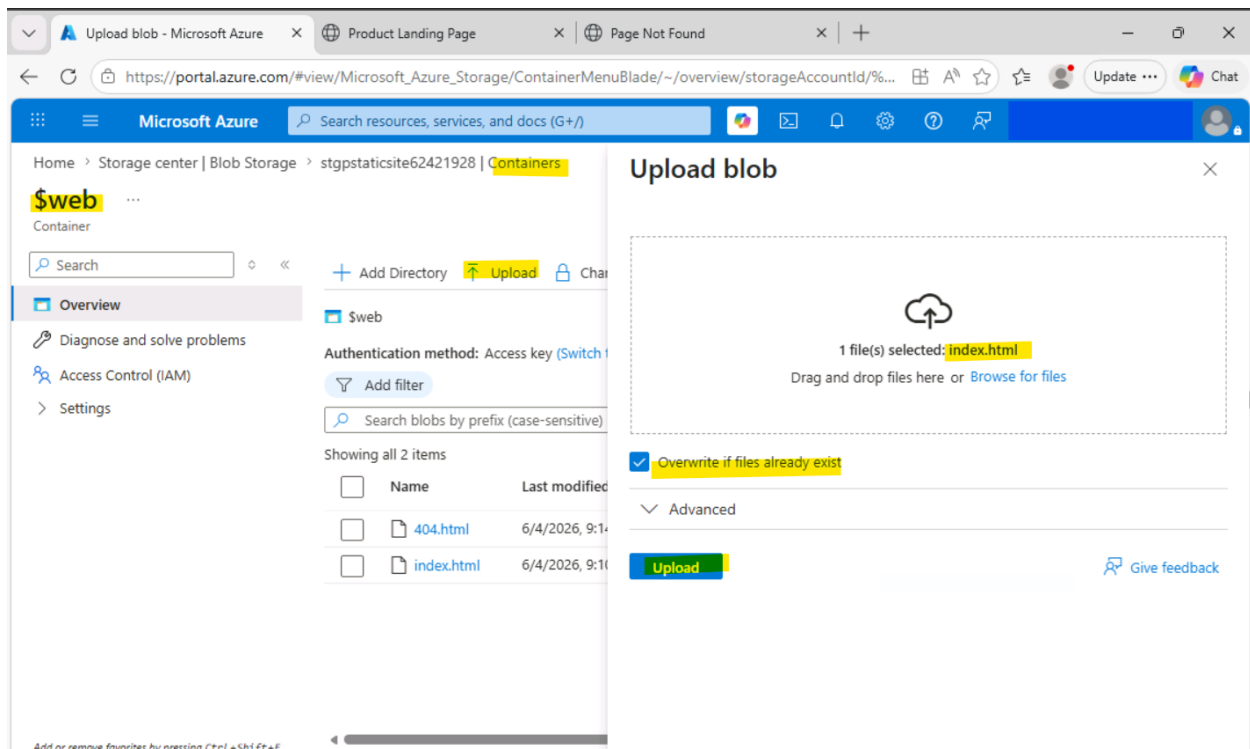
```
<!DOCTYPE html>
<html>
<head><title>Product Landing Page</title></head>
<body>
  <h1>Version 2 - Landing Page</h1>
  <p>Welcome to our updated product page with the latest information.</p>
</body>
</html>
```

Create version 2 landing page html and overwriting the previous version.

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Upload the updated file

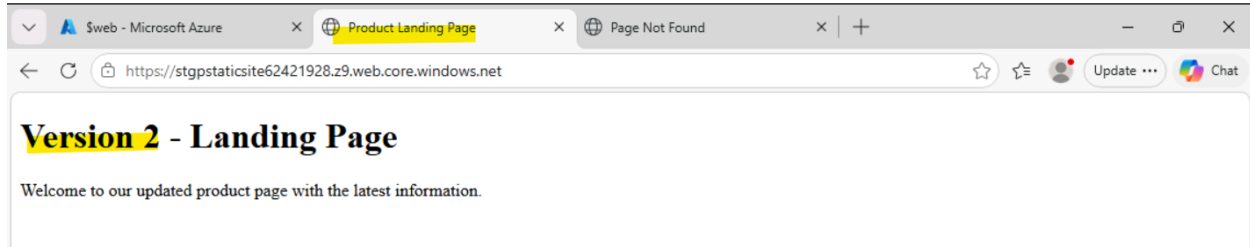
Replace the existing file in the \$web container with your updated version. Uploading a file with the same name overwrites the previous version, which is how you publish content updates to a static website.



Upload the updated file and make sure to check the 'Overwrite if files already exist'.

Verify the updated site

Confirm that the live website now shows the updated content. Because static website hosting serves files directly from blob storage, the change should appear immediately.

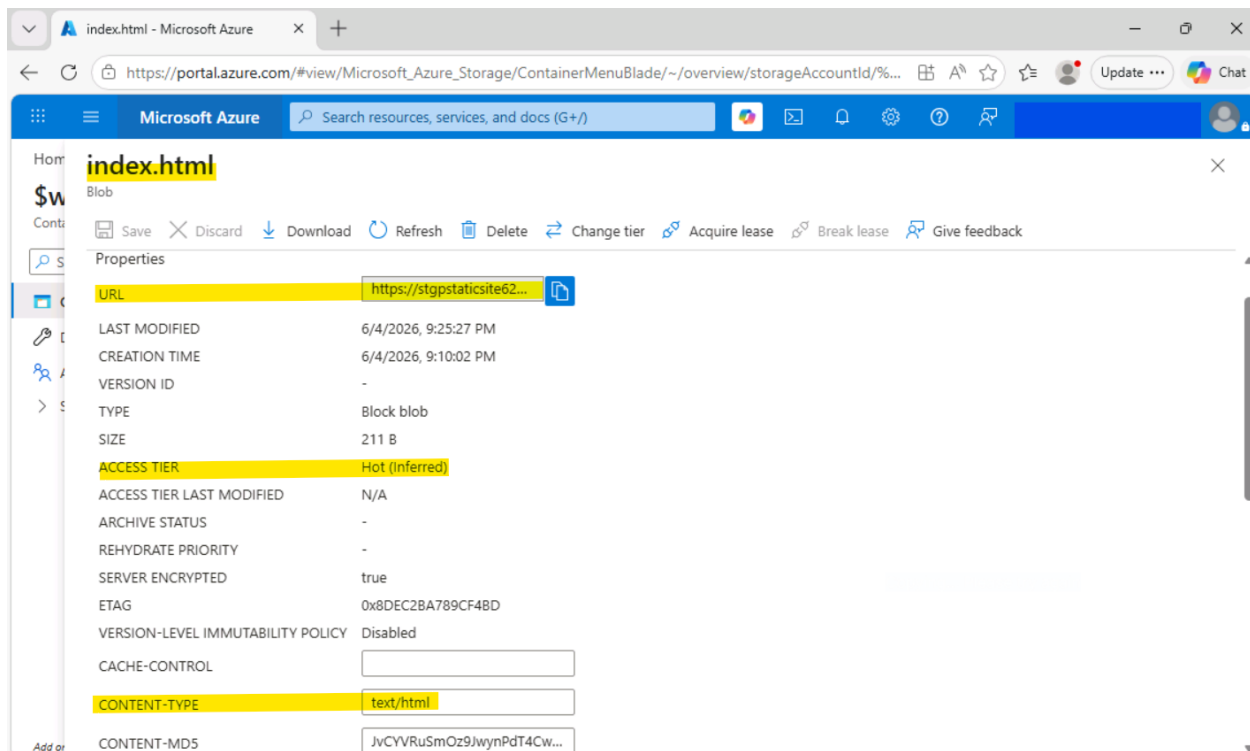


Refresh the Product Landing Page to verify the update. Now the live site shows **Version 2 – Landing Page** and the updated message.

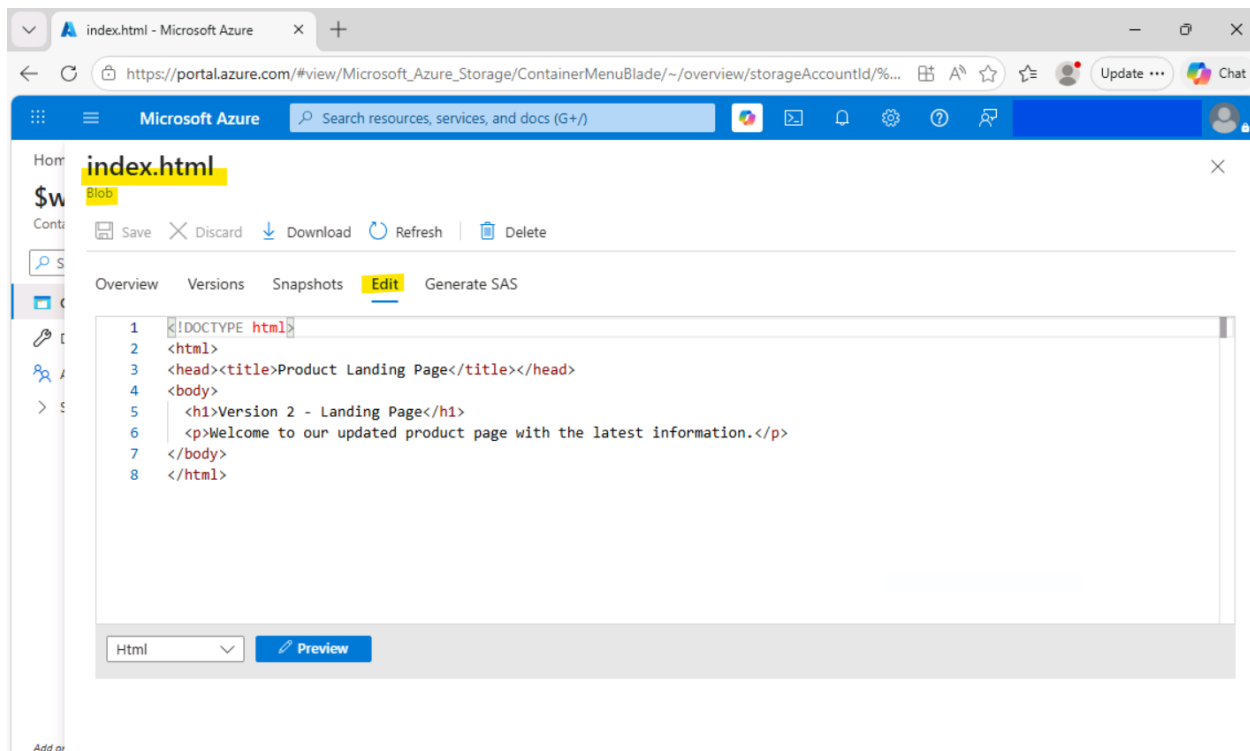
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Review blob properties and access tier

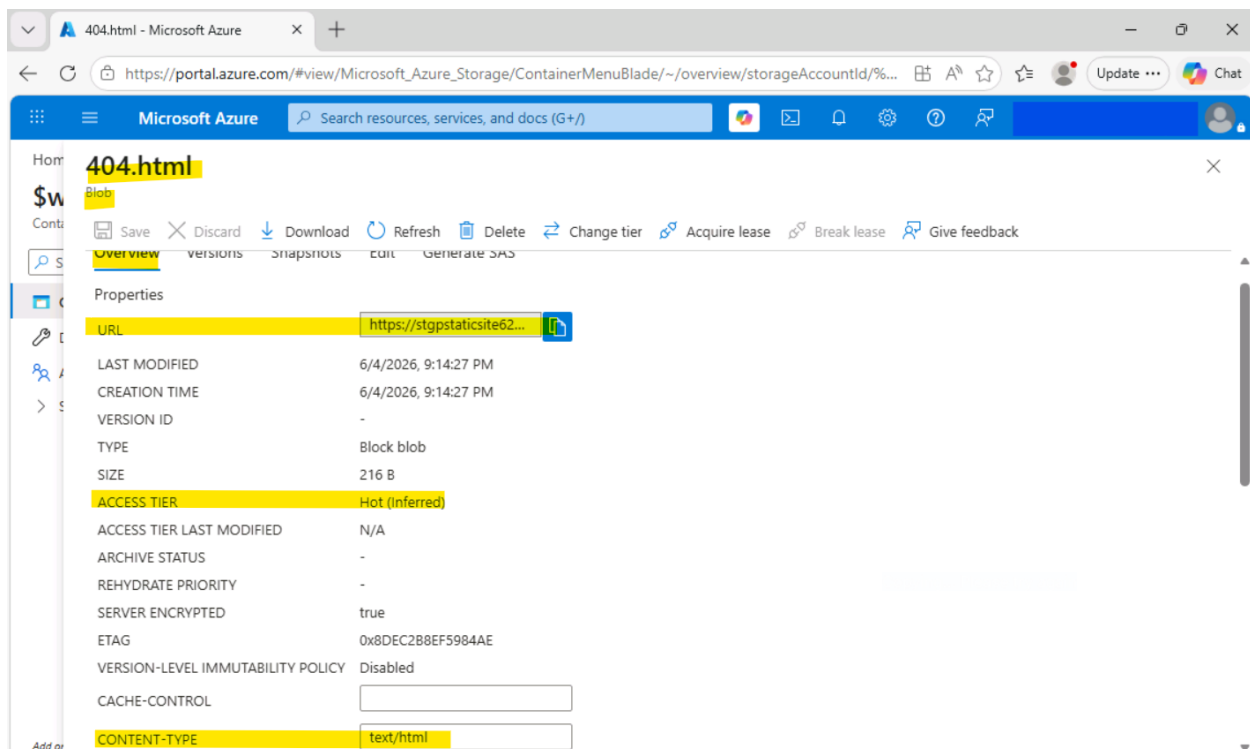
Examine the properties of your uploaded files to understand how Azure stores them. Each blob has metadata, an access tier, and a content type that affects how browsers handle it.



1. In the Overview of index.html blob, check the URL, Access Tier, and Content type.



2. Under Edit tab in the index.html blob, review the raw content to confirm it matches your Version 2 HTML.



3. In the Overview of 404.html blob, compare the URL, Access Tier, and Content type with index.html blob. Both blobs show **Content type** of **text/html** and **Access tier** of **Hot**.

Estimated cost using Azure Pricing Calculator

To demonstrate the cost-effectiveness of Azure Static Website Hosting, the Azure Pricing Calculator was used to estimate monthly costs for two common deployment scenarios: a personal portfolio website and a medium-sized company landing page. These estimates illustrate how Azure Blob Storage can provide a low-cost hosting solution for static web content without requiring virtual machines, databases, or web application services.

Actual costs may vary depending on storage usage, transactions, and outbound data transfer.

- **Situation: Small Personal Portfolio** (approx. CA\$0.26/month)
 - 1 Resource Group
 - 1 Storage Account (< 1GB)
 - 6 small HTML files with a few screenshots
 - Introduction
 - Resume
 - About Me page
 - Projects page
 - Contact Page
 - 404 error page
 - Static Website Hosting
 - No VMs, databases, or App Service
- Estimate numbers of Metrics
 - 100 visitors
 - 500 reads
 - 10 writes
- Service: approx. CA\$0.26/month
 - Azure Storage Account (1GB, Standard LRS): ~ CA\$0.03
 - Transactions & Data Transfer: ~ CA\$0.23

- **Situation: Med-size Company landing page** (approx. CA\$4.53/month)
 - 1 Storage Account (< 10GB)
 - HTML files
 - Homepage
 - About Us
 - Service
 - Careers
 - Contact Us
 - Product pages
 - Images and logos
 - Static Website Hosting
 - No VMs, databases, or App Service
- Estimate numbers of Metrics
 - 10,000 to 20,000 visitors
 - 500 Read Operations
 - 10 Write Operations
 - 5 List/Create Operations
 - 10 Other Operations
- Service: approx. CA\$4.53/month
 - Azure Storage Account (1GB, Standard LRS): ~ CA\$0.28
 - Transactions & Data Transfer: ~ CA\$4.25